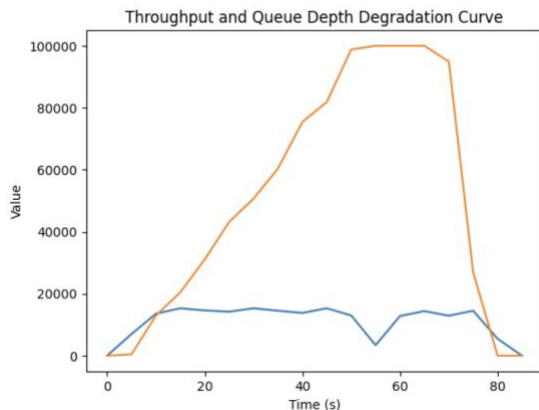


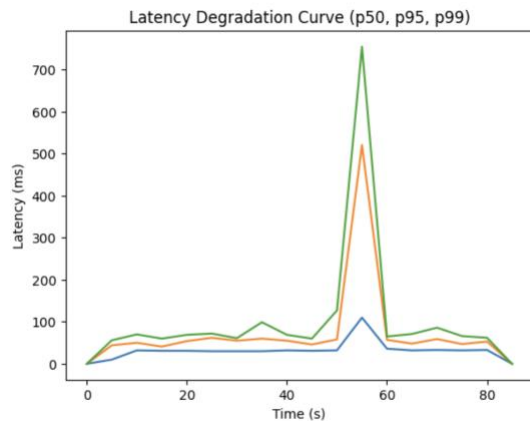
Notes for this test run: For my initial tests, not all 1,000,000 messages were written to the database because the DatabaseWriter could not keep up with the incoming message rate, causing the MessageBuffer to fill up to its maximum capacity of 100,000. Once full, new messages were dropped via offer(). To fix this, I made the consumer thread to be blocked until buffer space became available (using offer(msg, 2000ms) in MessageProcessingService). With back-pressure in place, all 1,000,000 messages were successfully written to the database. As shown in the screenshot, once the queue depth reached 100,000 (the configured maximum), DB write throughput dropped significantly and batch write latency (p50/p95/p99) spiked. This is evidence of back-pressure in action: consumer threads were blocked waiting for buffer space, which slowed down RabbitMQ ACKs and throttled message intake. Once the DatabaseWriter drained enough of the buffer, throughput recovered and latency returned to normal. This also explained the throughput in client side is lower than the 500,000 load test. I have also included screenshots for the failed test at the very end of this document.

```
[DB METRICS] dbThroughput=0/sec totalWritten=0 queueDepth=0 p50=0ms p95=0ms p99=0ms
[DB METRICS] dbThroughput=6993/sec totalWritten=34969 queueDepth=455 p50=10ms p95=44ms p99=56ms
[DB METRICS] dbThroughput=13549/sec totalWritten=102714 queueDepth=12950 p50=32ms p95=50ms p99=70ms
[DB METRICS] dbThroughput=15300/sec totalWritten=179214 queueDepth=20472 p50=31ms p95=41ms p99=60ms
[DB METRICS] dbThroughput=14600/sec totalWritten=252214 queueDepth=31067 p50=31ms p95=54ms p99=69ms
[DB METRICS] dbThroughput=14200/sec totalWritten=323214 queueDepth=43137 p50=30ms p95=62ms p99=72ms
[DB METRICS] dbThroughput=15300/sec totalWritten=399714 queueDepth=50678 p50=30ms p95=55ms p99=61ms
[DB METRICS] dbThroughput=14500/sec totalWritten=472214 queueDepth=60395 p50=30ms p95=60ms p99=99ms
[DB METRICS] dbThroughput=13800/sec totalWritten=541214 queueDepth=75413 p50=32ms p95=55ms p99=69ms
[DB METRICS] dbThroughput=15300/sec totalWritten=617714 queueDepth=81854 p50=31ms p95=46ms p99=60ms
[DB METRICS] dbThroughput=13000/sec totalWritten=682714 queueDepth=98784 p50=32ms p95=58ms p99=127ms
[DB METRICS] dbThroughput=3400/sec totalWritten=699714 queueDepth=100000 p50=110ms p95=521ms p99=755ms
[DB METRICS] dbThroughput=12800/sec totalWritten=763714 queueDepth=100000 p50=36ms p95=57ms p99=65ms
[DB METRICS] dbThroughput=14400/sec totalWritten=835714 queueDepth=100000 p50=32ms p95=48ms p99=71ms
[DB METRICS] dbThroughput=12900/sec totalWritten=900214 queueDepth=94914 p50=33ms p95=59ms p99=86ms
[DB METRICS] dbThroughput=14500/sec totalWritten=972714 queueDepth=26786 p50=32ms p95=47ms p99=66ms
[DB METRICS] dbThroughput=5457/sec totalWritten=1000000 queueDepth=0 p50=33ms p95=53ms p99=62ms
[DB METRICS] dbThroughput=0/sec totalWritten=1000000 queueDepth=0 p50=0ms p95=0ms p99=0ms
[DB METRICS] dbThroughput=0/sec totalWritten=1000000 queueDepth=0 p50=0ms p95=0ms p99=0ms
```



Name: Aijing Huang

CS6650: Scalable Distributed Systems – Assignment 3



Below shows no message drop

```
[ec2-user@ip-172-31-21-86 ~]$ grep "DB buffer full" consumer.log | wc -l
0
```

Test Run Successfully: all 1,000,000 messages broadcasted successfully

====Load test completed====

=== Warmup Phase Performance Metrics With 32 Threads ===

Messages sent: 32000

Successful Message: 32000

Failed Messages: 0

Duration: 4881 ms

Throughput: 6556.03 msg/sec

Success rate: 100.00%

=== Main Phase Performance Metrics With 16 Threads ===

Messages sent: 968000

Successful Message: 968000

Failed Messages: 0

Duration: 69176 ms

Throughput: 13993.29 msg/sec

Success rate: 100.00%

=== Overall Performance Metrics With 16 Threads ===

Messages sent: 1000000

Successful Message: 1000000

Failed Messages: 0

Duration: 74198 ms

Throughput: 13477.45 msg/sec

Success rate: 100.00%

Total Connections: 20

Total Reconnections: 0

Generating detailed statistical analysis...

Processing CSV data...

Metrics automatically pulled after test: all 1,000,000 written to DB (see last line in screenshot)

Name: Aijing Huang
CS6650: Scalable Distributed Systems – Assignment 3

```
=====
Fetching metrics from: http://35.94.24.99:8081/metrics
=====
[METRICS] Response:
{
  "messageStats": {
    "total_messages": 1000000,
    "earliest_timestamp": "2026-03-31T20:17:10.512338032Z",
    "latest_timestamp": "2026-03-31T20:18:17.161724773Z",
    "messages_per_sec": 15004,
    "messages_per_min": 900239,
    "duration_ms": 66649
  },
  "topUsers": [
    {
      "user_id": "90365",
      "message_count": 27
    },
    {
      "user_id": "19507",
      "message_count": 27
    },
    {
      "user_id": "23459",
      "message_count": 27
    },
    {
      "user_id": "50462",
      "message_count": 26
    },
    {
      "user_id": "96307",
      "message_count": 26
    }
  ],
  "topRooms": [
    {
      "room_id": "7",
      "message_count": 50430
    },
    {
      "room_id": "10",
      "message_count": 50396
    },
    {
      "room_id": "20",
      "message_count": 50313
    },
    {
      "room_id": "9",
      "message_count": 50311
    },
    {
      "room_id": "17",
      "message_count": 50090
    }
  ],
  "userParticipationPatterns": [
    {
      "user_id": "92437",
      "rooms_count": 17,
      "total_messages": 21
    },
    {
      "user_id": "77773",
      "rooms_count": 17,
      "total_messages": 19
    },
    {
      "user_id": "25191",
      "rooms_count": 16,
      "total_messages": 18
    },
    {
      "user_id": "18480",
      "rooms_count": 16,
      "total_messages": 23
    },
    {
      "user_id": "15440",
      "rooms_count": 16,
      "total_messages": 17
    }
  ],
  "totalMessages": 1000000
}
```

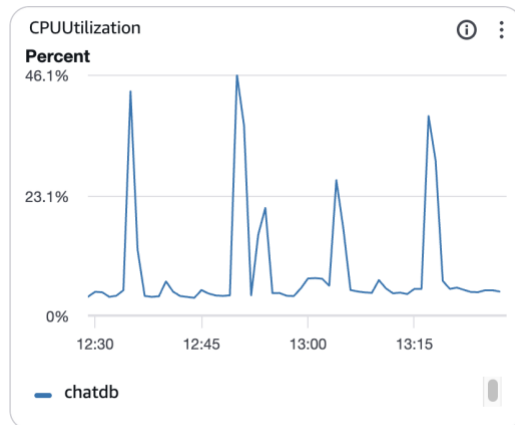
Write throughput, Write latency, Queue Depth

Queue depth: Queue was full during the test.

```
[DB METRICS] dbThroughput=0/sec totalWritten=0 queueDepth=0 p50=0ms p95=0ms p99=0ms
[DB METRICS] dbThroughput=6993/sec totalWritten=34969 queueDepth=455 p50=10ms p95=44ms p99=56ms
[DB METRICS] dbThroughput=13549/sec totalWritten=102714 queueDepth=12950 p50=32ms p95=50ms p99=70ms
[DB METRICS] dbThroughput=15300/sec totalWritten=179214 queueDepth=20472 p50=31ms p95=41ms p99=60ms
[DB METRICS] dbThroughput=14600/sec totalWritten=252214 queueDepth=31067 p50=31ms p95=54ms p99=69ms
[DB METRICS] dbThroughput=14200/sec totalWritten=323214 queueDepth=43137 p50=30ms p95=62ms p99=72ms
[DB METRICS] dbThroughput=15300/sec totalWritten=399714 queueDepth=50678 p50=30ms p95=55ms p99=61ms
[DB METRICS] dbThroughput=14500/sec totalWritten=472214 queueDepth=60395 p50=30ms p95=60ms p99=99ms
[DB METRICS] dbThroughput=13800/sec totalWritten=541214 queueDepth=75413 p50=32ms p95=55ms p99=69ms
[DB METRICS] dbThroughput=15300/sec totalWritten=617714 queueDepth=81854 p50=31ms p95=46ms p99=60ms
[DB METRICS] dbThroughput=13000/sec totalWritten=682714 queueDepth=98784 p50=32ms p95=58ms p99=127ms
[DB METRICS] dbThroughput=3400/sec totalWritten=699714 queueDepth=100000 p50=110ms p95=521ms p99=755ms
[DB METRICS] dbThroughput=12800/sec totalWritten=763714 queueDepth=100000 p50=36ms p95=57ms p99=65ms
[DB METRICS] dbThroughput=14400/sec totalWritten=835714 queueDepth=100000 p50=32ms p95=48ms p99=71ms
[DB METRICS] dbThroughput=12900/sec totalWritten=900214 queueDepth=94914 p50=33ms p95=59ms p99=86ms
[DB METRICS] dbThroughput=14500/sec totalWritten=972714 queueDepth=26786 p50=32ms p95=47ms p99=66ms
[DB METRICS] dbThroughput=5457/sec totalWritten=1000000 queueDepth=0 p50=33ms p95=53ms p99=62ms
[DB METRICS] dbThroughput=0/sec totalWritten=1000000 queueDepth=0 p50=0ms p95=0ms p99=0ms
[DB METRICS] dbThroughput=0/sec totalWritten=1000000 queueDepth=0 p50=0ms p95=0ms p99=0ms
```

Test was run around 13:16

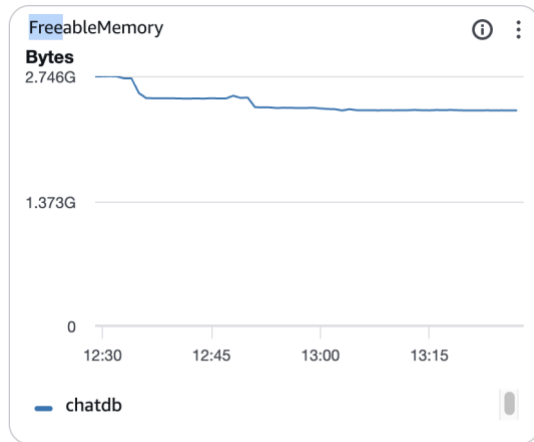
Database CPU



Database Memory: since RDS provides data for freeableMemory and the freeable memory barely change during the test run, the test barely use any memory.

Name: Aijing Huang

CS6650: Scalable Distributed Systems – Assignment 3



Failed Test Stats: Higher throughput but around 100,000 messages got dropped

====Load test completed====

=== Warmup Phase Performance Metrics With 32 Threads ===

Messages sent: 32000
Successful Message: 32000
Failed Messages: 0
Duration: 3366 ms
Throughput: 9506.83 msg/sec
Success rate: 100.00%

=== Main Phase Performance Metrics With 16 Threads ===

Messages sent: 968000
Successful Message: 968000
Failed Messages: 0
Duration: 54173 ms
Throughput: 17868.68 msg/sec
Success rate: 100.00%

=== Overall Performance Metrics With 16 Threads ===

Messages sent: 1000000
Successful Message: 1000000
Failed Messages: 0
Duration: 57668 ms
Throughput: 17340.64 msg/sec
Success rate: 100.00%
Total Connections: 20
Total Reconnections: 0
Generating detailed statistical analysis...
Processing CSV data...

Name: Aijing Huang
CS6650: Scalable Distributed Systems – Assignment 3

```
=====
Fetching metrics from: http://54.212.194.200:8081/metrics
=====
```

```
[METRICS] Response:
{
  "messageStats": {
    "total_messages": 927627,
    "earliest_timestamp": "2026-03-27T17:15:56.411238916Z",
    "latest_timestamp": "2026-03-27T17:16:52.382587510Z",
    "messages_per_sec": 16573,
    "messages_per_min": 994401,
    "duration_ms": 55971
  },
  "topUsers": [
    {
      "user_id": "94592",
      "message_count": 25
    },
    {
      "user_id": "75850",
      "message_count": 25
    },
    {
      "user_id": "35826",
      "message_count": 25
    },
    {
      "user_id": "80120",
      "message_count": 24
    },
    {
      "user_id": "51592",
      "message_count": 24
    }
  ],
  "topRooms": [
    {
      "room_id": "5",
      "message_count": 47452
    },
    {
      "room_id": "14",
      "message_count": 47409
    },
    {
      "room_id": "11",
      "message_count": 47145
    },
    {
      "room_id": "8",
      "message_count": 47136
    },
    {
      "room_id": "6",
      "message_count": 47088
    }
  ],
  "userParticipationPatterns": [
    {
      "user_id": "38972",
      "rooms_count": 17,
      "total_messages": 19
    },
    {
      "user_id": "68649",
      "rooms_count": 17,
      "total_messages": 22
    },
    {
      "user_id": "46930",
      "rooms_count": 16,
      "total_messages": 20
    },
    {
      "user_id": "26504",
      "rooms_count": 16,
      "total_messages": 20
    },
    {
      "user_id": "45612",
      "rooms_count": 16,
      "total_messages": 21
    }
  ],
  "totalMessages": 927627
}
```

Name: Aijing Huang

CS6650: Scalable Distributed Systems – Assignment 3

```
[DB METRICS] dbThroughput=0/sec totalWritten=500000 queueDepth=0
[DB METRICS] dbThroughput=64/sec totalWritten=500320 queueDepth=0
[DB METRICS] dbThroughput=13054/sec totalWritten=565593 queueDepth=4149
[DB METRICS] dbThroughput=16500/sec totalWritten=648093 queueDepth=10733
[DB METRICS] dbThroughput=15000/sec totalWritten=723093 queueDepth=26584
[DB METRICS] dbThroughput=15900/sec totalWritten=802593 queueDepth=37742
[DB METRICS] dbThroughput=15600/sec totalWritten=880593 queueDepth=50804
[DB METRICS] dbThroughput=14900/sec totalWritten=955093 queueDepth=66030
[DB METRICS] dbThroughput=14600/sec totalWritten=1028093 queueDepth=84326
[DB METRICS] dbThroughput=14800/sec totalWritten=1102093 queueDepth=99548
[DB METRICS] dbThroughput=13000/sec totalWritten=1167093 queueDepth=99563
[DB METRICS] dbThroughput=12900/sec totalWritten=1231593 queueDepth=99548
[DB METRICS] dbThroughput=14300/sec totalWritten=1303093 queueDepth=98970
[DB METRICS] dbThroughput=14000/sec totalWritten=1373093 queueDepth=54034
[DB METRICS] dbThroughput=10906/sec totalWritten=1427627 queueDepth=0
[DB METRICS] dbThroughput=0/sec totalWritten=1427627 queueDepth=0
```